

Emergence of Spinorial Structure from Chiral Vacuum Transport with Explicit $SU(2)$ Double-Cover Holonomy

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Abstract

This note explores a geometric route by which spinorial matter structure may arise within a Global Constraint Framework (GCF). In standard relativistic quantum theory, spinors enter through the representation theory of the Lorentz group, with the familiar double-cover property that a spinorial state changes sign under a 2π rotation and returns to itself only after a 4π rotation. Here we investigate whether this topological signature can be motivated by coherent transport through a structured chiral vacuum.

The proposal is deliberately conservative. We do not claim a complete derivation of fermionic quantum field theory or the spin-statistics theorem. Instead, we show that if the admissible vacuum has both a discrete structural baseline and a continuous chiral twist, then local state transport is naturally represented by a two-component object acted on by an $SU(2)$ transport holonomy. The minimal double-cover transport operator is

$$U(\phi) = \exp\left(-\frac{i}{2}\phi \mathbf{n} \cdot \boldsymbol{\sigma}\right),$$

where $\boldsymbol{\sigma}$ are the Pauli matrices. This gives

$$U(2\pi) = -I, \quad U(4\pi) = +I,$$

which is precisely the spinorial double-cover signature.

In this formulation, the two-component Weyl spinor is interpreted as the minimal transport object required to track chiral vacuum orientation. Relativistic mass then couples left- and right-chiral transport sectors, yielding the four-component Dirac object. The result is not a replacement for the standard Lorentz-group derivation of spinors, but a geometric reconstruction showing how spinorial topology is naturally compatible with, and potentially motivated by, chiral vacuum admissibility.

1 Motivation

Previous work within the Global Constraint Framework has shown that if admissible matter evolution is required to preserve coherent phase transport, and if one further imposes first-order relativistic closure, then the Dirac equation arises as the natural linear relativistic

matter equation. In that route, the Clifford algebra follows from the requirement that a first-order Hamiltonian square to the relativistic dispersion relation,

$$E^2 = p^2 + m^2.$$

That argument explains why the Dirac form is structurally natural once a first-order coherent matter equation has been assumed. However, it leaves open a deeper geometric question:

Why should the transported matter object be spinorial in the first place?

In standard quantum theory, spinors are introduced through the representation theory of rotations and the Lorentz group. This is mathematically rigorous, but it treats the spinor primarily as an abstract representation-theoretic object. The purpose of the present note is to provide a complementary geometric motivation: spinorial structure may be understood as the minimal transport structure required for coherent propagation through a vacuum possessing chiral topology.

We therefore ask whether the 4π return property of spinors can be interpreted as the macroscopic signature of transport through a bipartite, chiral admissibility geometry.

2 Structured Vacuum and Bipartite Transport

In a featureless scalar vacuum, a local excitation can be described by a single phase-like degree of freedom. Transport from one point to another is then trivial up to an ordinary scalar phase.

However, in a structurally closed vacuum, the background need not be featureless. In the Modal Theory / GCF setting, the admissible vacuum is naturally associated with:

1. a discrete structural baseline, such as a Z_3 or hexagonal phase skeleton;
2. a chiral asymmetry or twist, represented at the transport level by a Z_2 -type orientation distinction.

A local excitation moving through such a vacuum must track more than one datum. It must preserve coherent information about:

1. its phase relative to the structural baseline;
2. its orientation relative to the chiral twist.

A single real scalar amplitude is therefore not the most natural local transport object. The minimal object capable of tracking a binary chiral orientation while preserving coherent phase transport is a two-component complex vector.

We write this local transport object as

$$\psi = \begin{pmatrix} u \\ v \end{pmatrix}.$$

At this stage, ψ should be interpreted as a transport doublet rather than as a fully quantized fermionic field. The claim is not yet that all spinor quantum field theory has been derived, but that a chiral vacuum geometry naturally motivates a two-component state space.

3 The Minimal Chiral Transport Space

The two-component space carries the fundamental representation of $SU(2)$. The generators of this representation are the Pauli matrices,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k,$$

and

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}I.$$

The $SU(2)$ generators are therefore

$$J_i = \frac{1}{2}\sigma_i.$$

A rotation by angle ϕ about a unit axis $\mathbf{n}$ is represented on the transport doublet by

$$U(\phi, \mathbf{n}) = \exp(-i\phi \mathbf{n} \cdot \mathbf{J}) = \exp\left(-\frac{i}{2}\phi \mathbf{n} \cdot \boldsymbol{\sigma}\right).$$

This is the central mathematical object of the present note. It is the minimal holonomy operator for coherent transport in a two-component chiral state space.

4 Explicit Double-Cover Derivation

We now show explicitly that this transport law has the spinorial 4π return property.

Let

$$N \equiv \mathbf{n} \cdot \boldsymbol{\sigma}.$$

Using the Pauli algebra and $\mathbf{n} \cdot \mathbf{n} = 1$, one has

$$N^2 = I.$$

Therefore the exponential can be evaluated exactly:

$$U(\phi, \mathbf{n}) = \exp\left(-\frac{i}{2}\phi N\right) = \cos\left(\frac{\phi}{2}\right) I - i \sin\left(\frac{\phi}{2}\right) N.$$

For a full 2π rotation,

$$U(2\pi, \mathbf{n}) = \cos(\pi)I - i \sin(\pi)N.$$

Since

$$\cos(\pi) = -1, \quad \sin(\pi) = 0,$$

we obtain

$$U(2\pi, \mathbf{n}) = -I.$$

Thus, after a 2π rotation, the two-component transport object does not return to its original state. It returns to its negative.

For a 4π rotation,

$$U(4\pi, \mathbf{n}) = \cos(2\pi)I - i\sin(2\pi)N.$$

Since

$$\cos(2\pi) = 1, \quad \sin(2\pi) = 0,$$

we obtain

$$U(4\pi, \mathbf{n}) = +I.$$

Therefore the state returns to itself only after a 4π rotation:

$$\psi \xrightarrow{2\pi} -\psi, \quad \psi \xrightarrow{4\pi} \psi.$$

This is exactly the defining double-cover behaviour of spinorial structure.

5 Interpretation as Vacuum Transport Holonomy

The above result is standard as a statement about $SU(2)$ spinors. The GCF interpretation adds a geometric reading.

If the vacuum contains a chiral transport distinction, then a local excitation must carry an internal orientation relative to that chiral structure. A physical 2π spatial rotation returns the excitation to the same external coordinate orientation, but the internal chiral transport frame has traversed only one sheet of the double-cover structure. The state therefore acquires a minus sign.

A second 2π rotation completes the internal chiral transport cycle. Only after 4π does the state return to both:

1. its original external spatial orientation;
2. its original internal chiral alignment.

The 4π return condition is therefore not treated here as an arbitrary algebraic oddity. It is interpreted as the holonomy signature of a two-sheeted chiral transport space.

In this language, the spinor is the minimal local state object whose transport is compatible with a vacuum that possesses chiral admissibility structure.

6 Relation to $SO(3)$ and the Double Cover

The ordinary three-dimensional rotation group is $SO(3)$. Its elements describe rotations of vectors in physical space. However, $SO(3)$ is not simply connected. Its simply connected double cover is $SU(2)$:

$$SU(2) \longrightarrow SO(3).$$

The map is two-to-one:

$$U \quad \text{and} \quad -U$$

correspond to the same physical rotation in $SO(3)$.

This means that a 2π rotation is already the identity in $SO(3)$, but it corresponds to $-I$ in $SU(2)$. A 4π rotation is required to return to $+I$ in $SU(2)$.

The present proposal is that a chiral vacuum transport structure naturally selects the double-cover description rather than the single-cover vector description. In other words, matter transport follows the topology of $SU(2)$ because the admissibility geometry of the vacuum is itself effectively double-sheeted at the chiral transport level.

7 Chiral Weyl Doublets

The two-component transport object can now be identified with the structure of a Weyl spinor.

A left-chiral doublet is written

$$\psi_L = \begin{pmatrix} u_L \\ v_L \end{pmatrix},$$

and a right-chiral doublet as

$$\psi_R = \begin{pmatrix} u_R \\ v_R \end{pmatrix}.$$

In the present interpretation, ψ_L and ψ_R represent transport objects aligned with opposite chiral orientations of the vacuum.

For a massless excitation, a single chiral sector can propagate independently. This is consistent with the standard Weyl equation, where left- and right-handed spinors decouple in the massless limit.

Thus, the Weyl spinor is interpreted as the minimal coherent transport object for a massless excitation moving along one chiral rail of the vacuum.

8 Mass and Coupling of Opposite Chiral Sectors

A massive excitation cannot remain confined to one chiral transport rail. Mass mixes left- and right-chiral sectors.

In standard relativistic spinor theory, the Dirac equation in chiral form couples two Weyl spinors:

$$\begin{aligned} i\sigma^\mu \partial_\mu \psi_R &= m\psi_L, \\ i\bar{\sigma}^\mu \partial_\mu \psi_L &= m\psi_R, \end{aligned}$$

where

$$\sigma^\mu = (I, \boldsymbol{\sigma}), \quad \bar{\sigma}^\mu = (I, -\boldsymbol{\sigma}).$$

The four-component Dirac spinor is then

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}.$$

In the GCF interpretation, mass represents the requirement that coherent propagation negotiate both chiral orientations of the vacuum. A massive object is therefore not described

by a single chiral doublet, but by the coupled pair of left- and right-handed transport doublets.

The four-component Dirac object is thus interpreted as the minimal transport structure for massive relativistic matter in a chiral admissibility geometry.

9 Connection to Dirac Closure

The previous GCF reconstruction of the Dirac equation proceeded from the following logic:

1. admissible matter evolution should preserve coherent phase transport;
2. coherent matter evolution motivates a first-order local evolution law;
3. relativistic closure requires the square of the Hamiltonian to reproduce

$$E^2 = p^2 + m^2;$$

4. the unique linear factorisation of this closure condition yields the Clifford algebra;
5. the covariant Dirac equation follows:

$$(i\gamma^\mu \partial_\mu - m)\Psi = 0.$$

The present note supplies a geometric complement to that argument. It explains why the local matter object entering the first-order relativistic equation is naturally spinorial: the chiral vacuum transport space is represented minimally by an $SU(2)$ doublet, and mass couples the two chiral doublets into a Dirac spinor.

Thus the two arguments fit together:

$$\text{chiral vacuum transport} \implies SU(2) \text{ doublet} \implies \text{Weyl spinor},$$

and

$$\text{massive relativistic closure} \implies \psi_L \oplus \psi_R \implies \text{Dirac spinor}.$$

10 Coupling to Gauge Transport

Once local phase comparison is nontrivial, ordinary derivatives must be replaced by covariant derivatives:

$$\partial_\mu \longrightarrow D_\mu.$$

For a $U(1)$ phase connection,

$$D_\mu = \partial_\mu + igA_\mu.$$

The Dirac equation becomes

$$(i\gamma^\mu D_\mu - m)\Psi = 0.$$

This provides the natural joining point between the matter-side transport argument and the gauge-connection argument. Spinorial structure describes the local chiral transport object, while the gauge connection describes the admissible comparison of local phase frames across spacetime.

Together they produce the standard low-energy Dirac–Maxwell structure:

$$(i\gamma^\mu D_\mu - m)\Psi = 0,$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The GCF interpretation is that both structures arise from the same underlying requirement: local physics must preserve global admissibility under coherent transport.

11 What Has and Has Not Been Shown

It is important to state the scope precisely.

What has been shown

- A chiral transport background naturally motivates a two-component local state object.
- The minimal two-component transport law is represented by an $SU(2)$ holonomy.
- The $SU(2)$ transport operator satisfies

$$U(2\pi) = -I, \quad U(4\pi) = +I,$$

giving the spinorial double-cover signature explicitly.

- A massless excitation may be represented by a single Weyl doublet.
- A massive excitation requires coupling of left- and right-chiral doublets, naturally giving a four-component Dirac object.
- The argument complements the previous relativistic-closure route to the Dirac equation.

What has not been shown

- A complete derivation of the spin-statistics theorem.
- A full quantisation of the transport doublet into fermionic creation and annihilation operators.
- A derivation of anticommutation relations from the vacuum topology alone.
- A unique microscopic derivation of the exact Z_2 chiral transport coordinate from a fully specified field configuration.

- A complete derivation of the full Standard Model fermion representation content.

Therefore this note should be read as a geometric reconstruction and motivation for spinorial topology, not as a complete derivation of fermionic quantum field theory.

12 Conclusion

The existence of spinorial matter can be given a natural geometric interpretation within a Global Constraint Framework.

If the vacuum is not a featureless background but a chiral admissibility structure, then local coherent transport requires a two-component object capable of tracking internal chiral orientation. The minimal transport holonomy on this doublet is an $SU(2)$ operator,

$$U(\phi, \mathbf{n}) = \exp\left(-\frac{i}{2}\phi \mathbf{n} \cdot \boldsymbol{\sigma}\right),$$

which immediately yields

$$U(2\pi) = -I, \quad U(4\pi) = +I.$$

This is the defining double-cover behaviour of spinors.

A massless excitation may remain in a single chiral transport sector, giving a Weyl spinor. A massive excitation must couple opposite chiral sectors, giving the four-component Dirac object. When combined with first-order relativistic closure, this leads naturally to the Dirac equation.

The result is modest but structurally significant. Spinorial structure need not be viewed only as an abstract algebraic input. It can also be interpreted as the transport signature of a chiral, bipartite vacuum geometry. In this sense, local matter structure, like gauge connection structure, may be understood as arising from the admissibility topology of the vacuum.